Simple Harmonic Motion - Lab Notes

Apparatus checklist

- Spring, mass set and metre rule
- Stopwatch readable to 0.01 s
- ◆ Two conducting spheres on insulated stands
- ☐ Signal generator and oscilloscope

Theory

Displacement is given by x = Asin(ωt + ϕ) ...(I)

Here ω is the angular frequency and ϕ the phase constant.

Relaxation period, 2τ = ρ m ne

Wavelength λ = 2π / k, and the decay constant is μ.

The ratio of surface charge densities (σ1 /σ2) is constant.

Scattering angle θ ≤ 30° for all runs.

Energy ∑E = √(2m) ⇒ conserved.

Navigate Schemas → HR_SCHEMA → Tables.

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